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Effectiveness of musculoskeletal manipulations in patients with neck pain: a systematic review and network meta-analysis

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ABSTRACT

Background Neck pain is a common global health problem and a leading cause of disability, imposing significant personal and societal burdens. Musculoskeletal manipulations are recommended as the first-line treatment for neck pain in clinical practice guidelines, valued for their non-invasive and cost-effectiveness.

Objective To determine the most effective musculoskeletal manipulation for neck pain by integrating both direct and indirect evidence through a network meta-analysis, enabling comparisons among interventions even in the absence of direct head-to-head evidence.

Design Systematic review and network meta-analysis.

Data sources Web of Science, the Cochrane Library, Embase, PubMed, Clinical Trials Registry, China National Knowledge Infrastructure, VIP Database and Wanfang Data were searched from January 2013 to May 2025.

Eligibility criteria Randomised controlled trials (RCTs) involving adults (aged ≥18 years) with neck pain receiving musculoskeletal manipulations.

Data extraction and synthesis Paired reviewers independently extracted data. The primary outcome was pain intensity, assessed using the visual analogue scale or numeric rating scale (with total scores of 10 or 100 points). Secondary outcomes included neck disability measured by the neck disability index (with total scores of 50 or 100 points), cervical range of motion (flexion and extension) measured in degrees using a universal goniometer and adverse events. All included studies were assessed for risk of bias using the RoB 2.0 tool and categorised as 'low', 'some concerns' or 'high'. A frequentist random-effects network meta-analysis was conducted to calculate weighted mean differences with its 95% confidence intervals were calculated. Comparison-adjusted funnel plots were used to assess publication bias when 10 or more studies were included, and the Grading of Recommendations, Assessment, Development and Evaluations approach was applied for the quality of evidence.

Results A total of 101 RCTs involving 7633 participants were included. Overall, 62 (61.3%) had high risk of bias, 18 (17.8%) had some concerns and 21 (20.7%) had low risk of bias. Moderate to very low-certainty evidence showed significant reduction in neck pain intensity compared with no treatment: multimodal treatment (pooled weighted mean difference (WMD): −36.65, 95% CI −61.02 to −12.28), active control (pooled WMD: −36.62, 95% CI −62.67 to −10.57), manual therapy (pooled WMD: −35.85, 95% CI −59.86 to −11.83), soft tissue technique (pooled WMD: −28.72, 95% CI −51.65 to −4.79) and mobilisation (pooled WMD: −20.23, 95% CI −39.87 to −0.06), with multimodal treatment being the most effective intervention (surface under the cumulative ranking curve (SUCRA)=100%). No publication bias was detected for neck pain intensity. Moreover, multimodal treatment was the most effective for reducing neck disability (SUCRA=96.3%), whereas manipulation was the most effective for improving cervical range of flexion (SUCRA=81.3%) and extension (SUCRA=78.2%).

Conclusion Multimodal treatment (the combined use of two or more musculoskeletal manipulations) is the most effective intervention for reducing both neck pain intensity and neck disability. Manipulation is the most effective for improving cervical range of motion.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This network meta-analysis systematically synthesised the available evidence to evaluate the comparative effectiveness of various musculoskeletal manipulations for adult neck pain.
⇒ This study, integrating both direct and indirect evidence, identified multimodal treatment as the most effective for neck pain and disability, and manipulation for cervical range of motion, providing robust evidence to inform clinical practice.
⇒ Risk of bias was assessed using the RoB 2.0 tool, revealing common methodological and design limitations in existing trials, providing valuable guidance for future research.
⇒ Subgroup analyses by neck pain chronology (eg, acute or chronic) were not possible due to the absence of relevant information in the included randomised controlled trials.
⇒ Long-term efficacy could not be assessed because of a lack of follow-up data, highlighting the need for further research on sustained treatment effects.

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INTRODUCTION

According to the Global Burden of Disease Study, neck pain refers to pain in the cervical spine region (with or without pain in the arms) that persists for ≥24 hours.[1] Owing to physical inactivity and increased daily computer use, neck pain has become a common complaint, affecting 288.7 million people globally. Its age-standardised prevalence rate is 27.0 per 1000 individuals, and this prevalence is expected to increase considerably in the future.[2] As the fourth leading cause of disability,[3] neck pain affects patients' social activities and daily living; it also imposes an economic burden.[4 5] Therefore, addressing neck pain is imperative.[6]

Several non-invasive interventions are recommended for neck pain, such as exercise, acupuncture and musculoskeletal manipulations.[7 8] Although guidelines prioritise exercise, musculoskeletal manipulations are associated with fewer adverse effects and lower costs than exercise.[9 10] Musculoskeletal manipulation includes a range of non-invasive physical treatment modalities applying targeted forces to articular structures and soft tissues, with the aim of facilitating immediate functional enhancement while promoting sustained tissue repair and remodelling. Current reviews of randomised controlled trials (RCTs) have supported the effectiveness of musculoskeletal manipulations,[11] making them widely adopted in clinical practice as the primary option.[12]

Various musculoskeletal manipulations are available, such as massage and manual therapy[13–16]; however, the most effective option, which is crucial for clinical decision-making, remains unclear. To compare the effectiveness of different musculoskeletal manipulations, a network meta-analysis (NMA) is required to rank all available interventions and identify the most effective option.[17] However, existing NMAs have only compared different categories of conservative interventions for neck pain, with the relative effectiveness of different types of musculoskeletal manipulation remaining unclear.[18]

Therefore, in this study, we systematically summarised the available evidence on musculoskeletal manipulations for neck pain and identified the most effective type of musculoskeletal manipulation to provide a reference work for clinical application.

METHODS

Study reporting

This NMA adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension Statement for Network Meta-Analyses (PRISMA-NMA).[19] The protocol was registered with PROSPERO (CRD42023420775) and has been previously published.[20] Ethics approval was not required for this review, as it is a literature review that does not involve direct contact with patients or the public, nor concerns related to their privacy.

Search strategy

Four English databases (Web of Science, Cochrane Central Register of Controlled Trials, Embase and PubMed) and three Chinese databases (China National Knowledge Infrastructure, VIP Database and Wanfang Data) were searched. The Clinical Trials Registry (clinicaltrials.gov) was also searched for unpublished trials. Initially, the publication timeframe of the studies was restricted to the period from 1 January 2013 to 30 April 2023. An updated search was performed on 29 May 2025 to include more recent publications. There were no language restrictions in our search strategy, though included studies were limited to English and Chinese due to practical screening constraints. The literature search strategies included terms for (i) musculoskeletal manipulations, (ii) neck pain and (iii) RCTs. The detailed search strategies are listed in online supplemental table 1).

Eligibility criteria

RCTs meeting the following criteria were included:

I. Participants: adults (aged ≥18 years) with neck pain localised to the neck region, defined as a primary complaint of pain between the occiput and the first thoracic vertebra.[8]
II. Interventions: participants were treated by a therapist using only musculoskeletal manipulations (eg, acupressure and mobilisation) of joints or soft tissues in the cervicothoracic region, regardless of the frequency or duration of treatment.
III. Comparison: any other musculoskeletal manipulations different from the intervention group, no treatment or active control (including education and self-exercise).
IV. Outcomes: the primary outcome was neck pain intensity (measured by different scales, such as the visual analogue scale (VAS) and numeric pain rating scale (NPRS)); the secondary outcomes included neck disability (measured by different scales, such as neck disability index (NDI)), cervical range of motion (flexion and extension measured using standardised devices) and adverse events.

The exclusion criteria were as follows:

I. Studies reporting primary complaints other than neck pain, such as postconcussion syndrome, headache and migraine.
II. RCTs published in neither Chinese nor English.

Study selection

Literature records were imported into the literature management software (PICO Portal, FL, United States, available at www.picoportal.org). After removing duplicate records, two reviewers independently screened the titles and abstracts. Subsequently, the full texts of potentially eligible studies were independently reviewed by the same two reviewers, with reasons for exclusion recorded.

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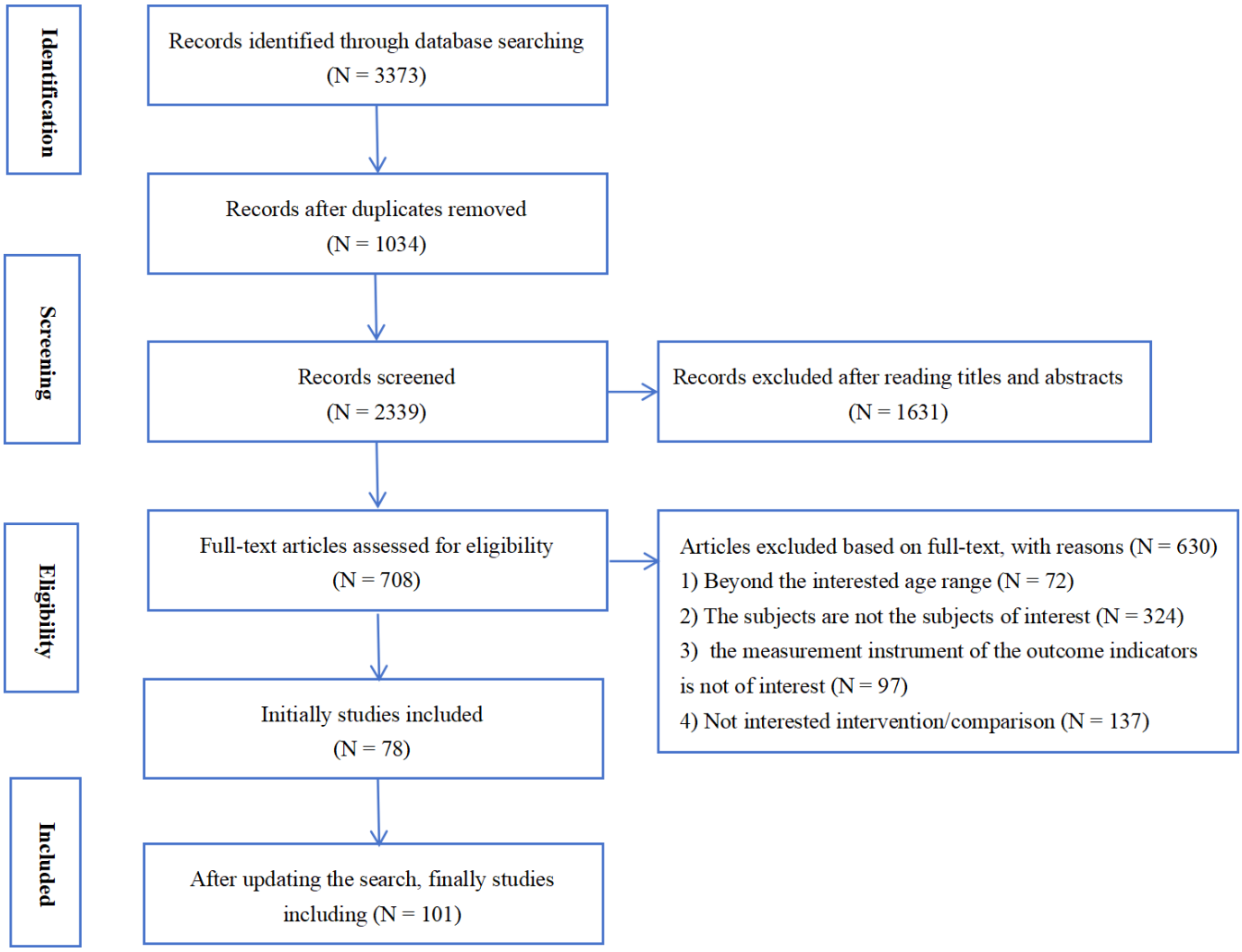


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram showing the literature search and selection.

Any disagreements were resolved through discussion or by consulting a third reviewer when necessary. Figure 1 shows the study selection process.

Data collection

Two reviewers independently extracted data from each study using a standardised extraction table, including (i) study identifiers (eg, first author, country or region and sample size); (ii) details of intervention and comparison (eg, components, frequency and duration); (iii) participant information (eg, type of neck pain and mean age) and (iv) outcomes (eg, neck pain intensity scale scores at pre- and postintervention, and measurements). Discrepancies were resolved by discussion to consensus.

Risk-of-bias assessment

Two reviewers independently evaluated the risk of bias for the included RCTs using the revised Cochrane risk-of-bias tool (RoB 2.0).[21] Any disagreements were resolved through discussion. When consensus could not be reached, a third reviewer was consulted. RoB 2.0 includes five domains: bias arising from the randomisation process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in the measurement of the outcome and bias in the selection of the reported result. Regarding the risk of bias, each domain was judged as 'low risk', 'some concerns' or 'high risk'. The overall risk of bias was assessed based on the results for each domain.

Statistical analysis

Effect estimates

We referred to a published review to categorise the identified specific interventions.[22] All included specific interventions were categorised into manipulation, manual therapy, soft tissue technique, mobilisation, multimodal treatment, active control and no treatment. Detailed definitions are provided in online supplemental table 2.

All statistical analyses were conducted using software STATA (V.16) and R (V.4.1.3). Neck pain intensity was measured using VAS or NPRS, with both scales employed in either 100- or 10-point formats across studies, while neck disability was assessed using the neck disability index (NDI) with different total scores (100 or 50). To facilitate

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pooled analysis, all means and SD were converted to a 100-point scale, and the weighted mean difference (WMD) with its 95% CI was reported. A WMD of ≥20 points for neck pain intensity and ≥15 points for neck disability was considered clinically meaningful.[23 24] For cervical range of motion (flexion and extension), angles were directly measured in degrees using standardised devices across studies. Therefore, the WMD with its 95% CI was also reported. Based on clinical experience, a WMD of ≥10 degrees was considered clinically meaningful. Safety outcomes were not quantitatively synthesised because most of the included RCTs did not report information regarding adverse events.

Pairwise meta-analysis

Before performing the pairwise meta-analyses, Cochran's Q and I-square (I^2) values were used to evaluate the heterogeneity qualitatively and quantitatively among the RCTs, respectively. For Cochran's Q test, p<0.1 indicated the absence of heterogeneity. I^2 values of <25%, 25–50% and >50% indicated low, moderate and high heterogeneity, respectively.[25] Given the possible heterogeneity among studies, a random-effects model was used to pool the effect estimates.[26]

Network meta-analysis

A random-effects model within a frequentist framework was used to assess the comparative effectiveness of eligible interventions.[27] The effect size for primary and secondary outcomes was reported as the WMD with 95% CI. Assumptions of transitivity and consistency were assessed using different methods.[28] Transitivity was evaluated by qualitatively comparing the clinical and methodological characteristics of the RCTs; when these characteristics were similar, transitivity was judged to be satisfactory.[29] Global inconsistency was detected using the inconsistency model,[30] p>0.05 indicated acceptable global consistency, and the consistency model was adopted; otherwise, the inconsistency model was applied. Local inconsistency was examined using the node-splitting method,[31] p<0.05 suggested significant inconsistency between direct and indirect comparisons, which may downgrade the certainty of evidence.[32 33]

All comparisons are displayed in a network plot, and the comparative effects of the interventions are shown in a league table. The surface under the cumulative ranking curve (SUCRA) showed the effectiveness ranking of the interventions for each outcome; a larger area indicated better effectiveness.[29] Publication bias was evaluated using a comparison-adjusted funnel plot and Egger's test when >10 RCTs were included.[34] The symmetry of the funnel plot points surrounding the zero line and p>0.05 in Egger's test indicated the absence of publication bias.

Subgroup analyses

Planned subgroup analyses by symptom duration (acute vs chronic) could not be performed as relevant data were inadequately reported in the included studies. Consequently, subgroup analysis was conducted based on the total scale range of the neck pain intensity measure: subgroup 1 included studies using scales with a total score of 100 points, and subgroup 2 included studies using scales with a total score of 10 points.

Sensitivity analyses

Initially, we planned to exclude RCTs with a high risk of bias for the sensitivity analysis of neck pain intensity. However, because more than half of the RCTs were assessed as having a high risk of bias, the sensitivity analysis was conducted by excluding RCTs with large differences in baseline scores (based on clinician judgement).

Certainty of evidence

The Grading of Recommendations, Assessment, Development and Evaluation tool for NMA was used to assess the certainty of evidence.[35 36] Two independent reviewers evaluated each outcome and disagreements were resolved via discussion or by consulting a third researcher. Initially, the certainty of evidence for direct comparisons was rated based on the risk of bias, inconsistency, publication bias and indirectness. Subsequently, the evidence quality for indirect comparisons was rated based on the dominant first-order loop and intransitivity. Finally, the evidence quality for network comparisons was rated as 'very low', 'low', 'moderate' or 'high', based on the higher rating between direct and indirect comparisons, considering incoherence and imprecision.

Patient and public involvement

None.

RESULTS

Study search and selection

In the initial study search, a total of 3373 records were identified from the 8 databases. After removing duplicates, 2339 records were subjected to eligibility assessment, and 1631 were excluded during the screening of the titles and abstracts. Subsequently, 708 records were subjected to full-text screening, of which a further 630 were excluded. After updating the search, 23 additional RCTs were included. Ultimately, 101 RCTs met the inclusion criteria. Figure 1 illustrates the detailed literature selection process.

Characteristics of the included RCTs

A total of 7633 participants were included in the 101 RCTs, with 80 (79.2%) RCTs having a sample size of ≤100, 20 (19.8%) having sample sizes ranging from 101 to 300 and only 1 (1.0%) having a sample size of >300. The mean age of participants ranged from 20 to 75 years. Most RCTs were conducted in Asia (n=65, 64.4%), followed by Europe (n=21, 20.8%), North America (n=11, 10.9%), South America (n=1, 1.0%), Africa (n=2, 1.9%) and Oceania (n=1, 1.0%). Most RCTs were conducted in hospital settings (n=57, 56.4%), followed by healthcare

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centres (n=15, 14.9%), private clinics (n=16, 15.8%), universities (n=12, 11.9%) and companies (n=1, 1.0%).

Multimodal treatment was the most commonly used intervention (n=38, 37.6%), followed by soft tissue technique (n=37, 36.6%), manipulation (n=26, 25.7%), active control (n=24, 23.8%), manual therapy (n=21, 20.8%) and mobilisation (n=17, 16.8%). The duration of intervention in most studies was no more than 4 weeks (n=79, 78.2%), followed by no more than 8 weeks (n=11, 10.9%), only nine studies (9.0) had a more than 8 weeks duration, with two studies (1.9%) not reporting the duration. All included 101 RCTs assessed neck pain intensity. Neck disability was reported in 63 (62.4%) studies, flexion in 22 (21.8%), extension in 23 (22.8%) and adverse events in 5 (5.0%) studies. The detailed trial characteristics are provided in online supplemental table 3.

Risk of bias

Inter-rater agreement was substantial (Cohen's kappa=0.52), with discrepancies resolved through discussion or third-reviewer consultation. Of the 101 included trials, 62 (61.3%) had a high overall risk of bias, 18 (17.8%) had some concerns about the possible risk of bias and 21 (20.7%) had a low overall risk of bias. The most common concerns involved the measurement of the outcome and the selection of the reported result. The measurement of neck pain intensity using a patient-reported scale resulted in 48 trials (47.5%) being assessed as having a high risk of bias regarding outcome measurement. Prior protocols were not provided by 52 trials (51.4%) and were judged as having some concerns regarding the selection of the reported results. However, most trials were considered to have a low risk of bias regarding missing outcome data (n=90, 89.1%) and deviations from intended interventions (n=75, 74.2%). Figure 2 presents the overall risk of bias of the included RCTs, and details are provided in online supplemental table 4.

Results of the pairwise meta-analysis

For neck pain intensity, compared with no treatment, soft tissue technique (pooled WMD: −27.63, 95% CI −38.62 to −1.64, I^2=88.8%, n=5), manipulation (pooled WMD: −13.86, 95% CI −22.48 to −5.24, I^2=64.1%, n=6) and manual therapy (pooled WMD: −22.33, 95% CI −28.08 to −16.57, I^2=60.5%, n=4) were associated with significant reductions. For neck disability, manipulation (pooled WMD: −13.57, 95% CI −16.87 to −10.27, I^2=0.0%, n=2) and manual therapy (pooled WMD: −12.30, 95% CI −17.36 to −7.24, I^2=82.1%, n=4) also showed significant improvements compared with no treatment. For the cervical range of motion, manipulation (flexion: WMD: 8.60, 95% CI 1.85 to 15.35, n=1; extension: WMD: 9.30, 95% CI 2.60 to 16.00, n=1), mobilisation (flexion: WMD: 5.10, 95% CI 0.50 to 9.70, n=1; extension: WMD: 6.50, 95% CI 0.90 to 12.10, n=1) and the soft tissue technique (flexion: pooled WMD: 9.85, 95% CI 4.16 to 15.55, I^2=0.0%, n=2; extension: pooled WMD: 8.39, 95% CI 3.37 to 13.40, I^2=0.0%, n=2) were found to be effective for both flexion and extension. The pairwise meta-analysis outcomes are shown in online supplemental table 5.

Results of the NMA

Neck pain intensity

Seven interventions from 68 trials were included in the network of comparisons (figure 3). Multimodal treatment was used in 31 studies (23.7%), followed by soft tissue techniques (n=27, 18.8%), manipulation (n=17, 11.8%), mobilisation (n=14, 9.7%) and manual therapy (n=13, 9.0%). Active control was the most frequently used comparator (n=24, 16.7%), while no treatment was used in 14 studies (9.7%). Neck pain intensity was assessed in all studies using patient-reported scales, with 14 studies employing a 100-point scale[37–50] and the remaining 54 studies[51–104] using a 10-point scale.

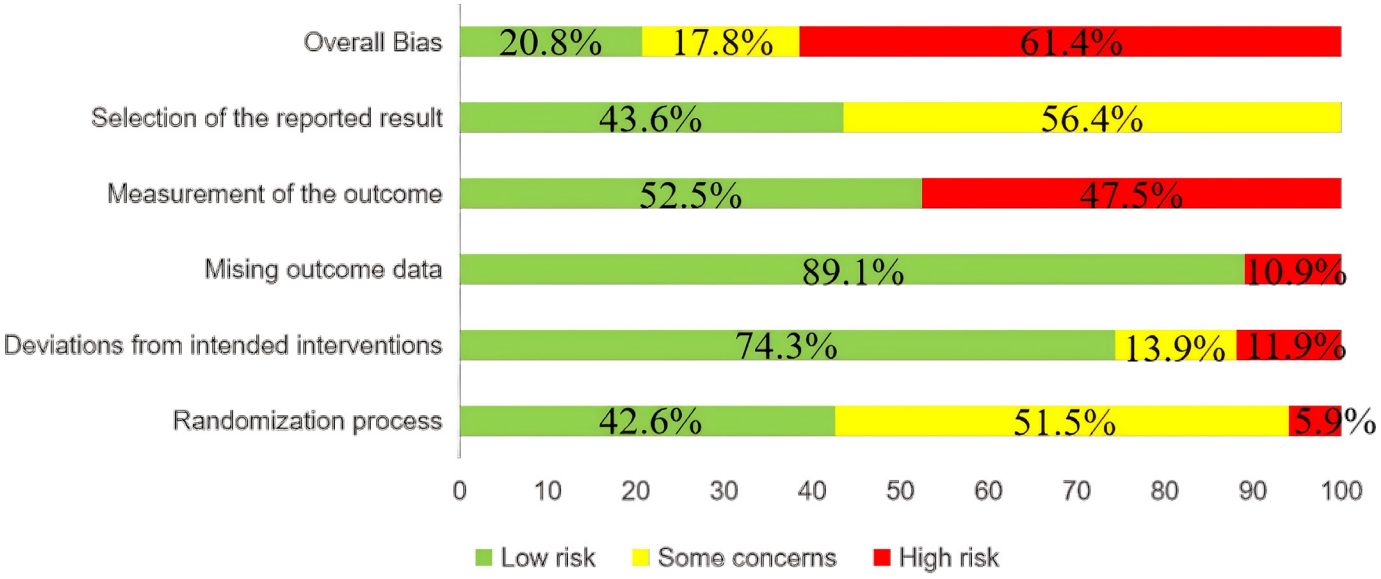


Figure 2 The overall risk of bias of all included randomised controlled trials.

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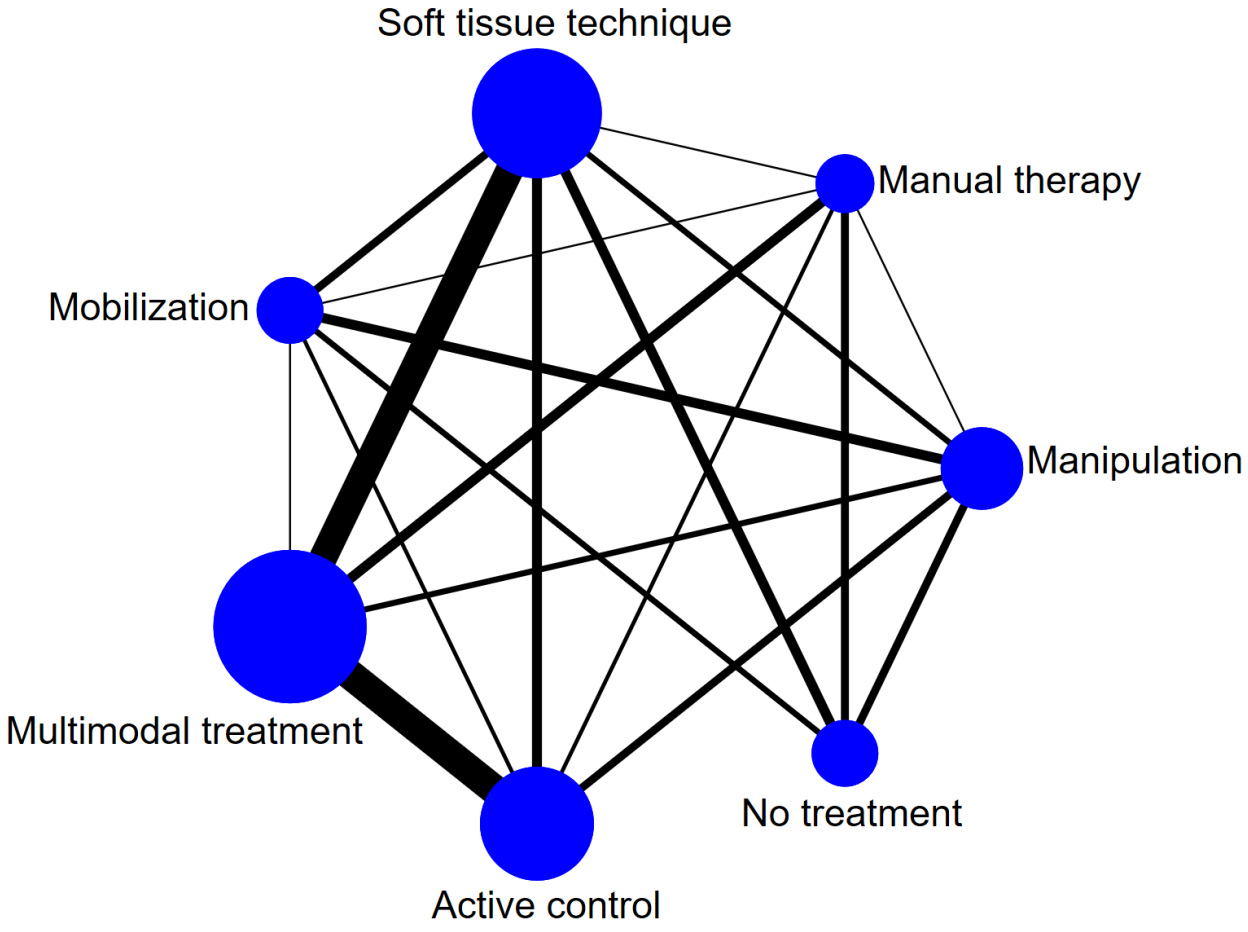


Figure 3 Network plot comparing the intensity of neck pain among different networks.

The inconsistency model was used due to the presence of global inconsistency (p=0.0078) (online supplemental table 8). The NMA results showed that, compared with no treatment, all musculoskeletal manipulations were significantly effective: multimodal treatment (pooled WMD: −36.65, 95% CI −61.02 to −12.28), active control (pooled WMD: −36.62, 95% CI −62.67 to −10.57), manual therapy (pooled WMD: −35.85, 95% CI −59.86 to −11.83), soft tissue technique (pooled WMD: −28.72, 95% CI −51.65 to −4.79), manipulation (pooled WMD: 16.61, 95% CI −6.40 to 39.63) and mobilisation (pooled WMD: −20.23, 95% CI −39.87 to −0.06). The SUCRA results showed that multimodal treatment (SUCRA=100%) had the highest probability of being the best option (figure 4). The quality of this evidence was judged to be moderate to very low (table 1).

Neck disability

Seven interventions from 40 trials were included in the network of comparisons (online supplemental figure 1). Multimodal treatment was used in 20 studies (23.8%), followed by soft tissue techniques (n=13, 15.4%), manual therapy (n=10, 11.9%), manipulation (n=9, 10.7%) and mobilisation (n=7, 8.3%). Active control was the most frequently used comparator (n=15, 17.8%), while no treatment was used in 10 studies (11.9%). Neck disability was assessed in all studies using patient-reported scales, with 14 studies employing a 100-point scale[39–42 51–60] and the remaining 26 studies[43–47 68–88] using a 50-point scale.

No significant global inconsistency (p=0.9949) or local inconsistency (p>0.05) was observed (online supplemental table 9). Then, the consistency model was adopted. The NMA results showed that, compared with no treatment, all musculoskeletal manipulations were significantly effective: multimodal treatment (pooled WMD: −18.22, 95% CI −23.88 to −12.57), mobilisation (pooled WMD: −15.10, 95% CI −21.83 to −8.37), active control (pooled WMD: −13.92, 95% CI −19.92 to −7.91), manipulation (pooled WMD: −12.90, 95% CI −18.89 to −6.90), soft tissue technique (pooled WMD: −12.35, 95% CI −17.67 to −7.03) and manual therapy (pooled WMD: −11.42, 95% CI −16.62 to −6.22). The SUCRA results showed that multimodal treatment (SUCRA=96.3%) had the highest probability of being the best option (online supplemental figure 2). The quality of this evidence was judged to be high to very low (online supplemental table 10).

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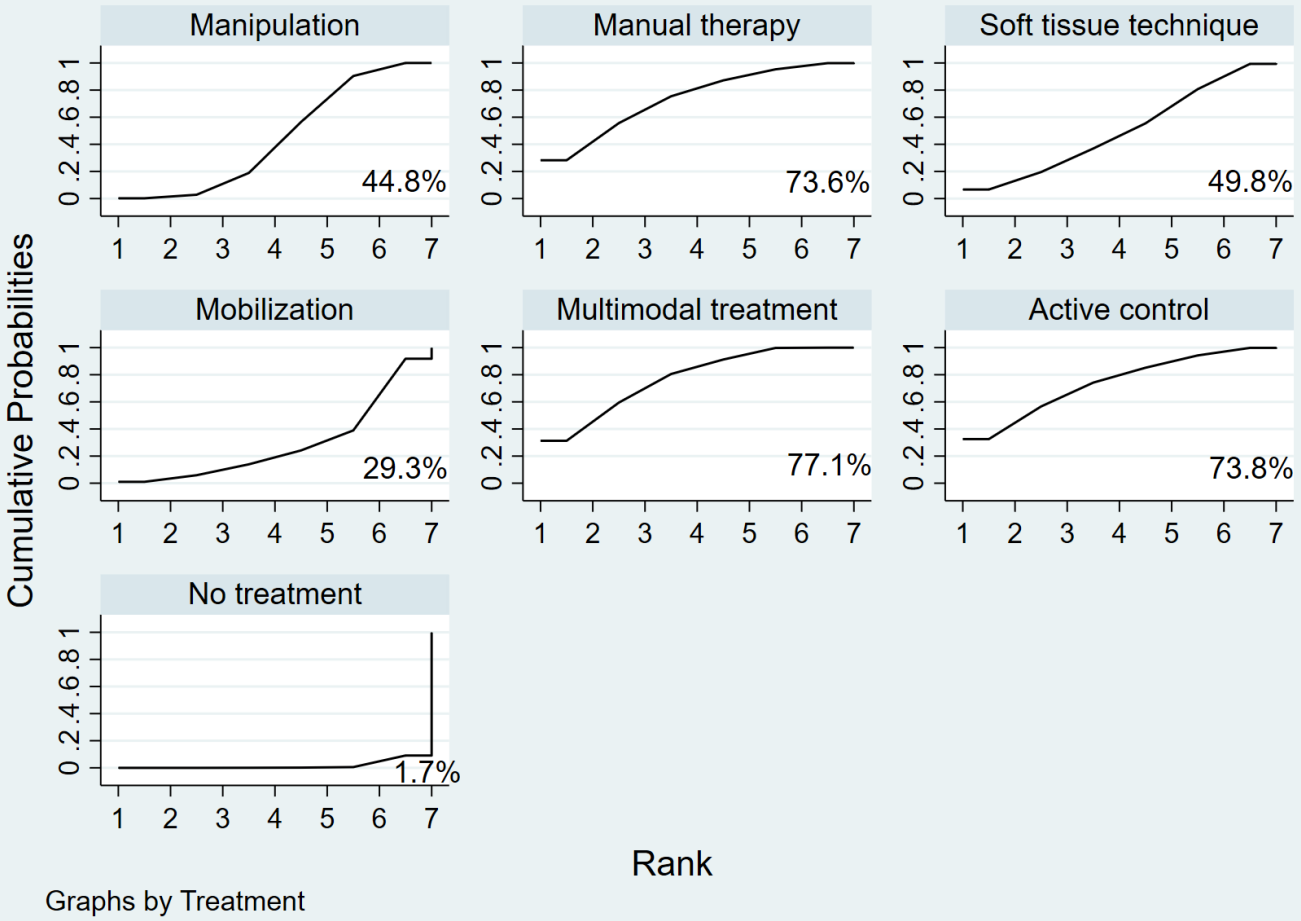


Figure 4 Surface under the cumulative ranking curve (SUCRA) for assessing the comparative effectiveness for reducing the intensity of neck pain.

Cervical range of motion

For flexion, 7 interventions from 20 trials were included (online supplemental figure 3). Multimodal treatment was used in 11 studies (25.0%), followed by soft tissue techniques (n=8, 18.1%), manipulation (n=5, 11.3%), manual therapy (n=4, 9.0%) and mobilisation (n=4, 9.0%). Active control was the most frequently used comparator (n=9, 20.4%), while no treatment was used in three studies (6.8%). Cervical range of motion was assessed in all studies using standardised devices. No significant global inconsistency (p=0.9530) or local inconsistencies (p>0.05) was observed (online supplemental table 11). Then, the consistency model was adopted. The NMA results showed that, compared with no treatment, manipulation (pooled WMD: 9.22, 95% CI 2.21 to 16.33), multimodal treatment (pooled WMD: 8.34, 95% CI 1.58 to 15.10), mobilisation (pooled WMD: 8.37, 95% CI 1.37 to 15.74) were significantly effective. The SUCRA results showed that manipulation (SUCRA=81.3%) had the highest probability of being the best treatment option (online supplemental figure 4). The quality of this evidence was judged to be moderate to very low (online supplemental table 12).

For extension, 7 interventions from 21 trials were included (online supplemental figure 5). Multimodal treatment was used in 12 studies (26.0%), followed by soft tissue techniques (n=8, 17.3%), manipulation (n=5, 10.8%), manual therapy (n=4, 8.6%) and mobilisation (n=4, 8.6%). Active control was the most frequently used comparator (n=10, 21.7%), while no treatment was used in three studies (6.5%). Cervical range of motion was assessed in all studies using standardised devices. No significant global inconsistency (p=0.9006) or local inconsistencies (p>0.05) was observed (online supplemental table 13). Then, the consistency model was adopted. The NMA results showed that, compared with no treatment, manipulation (pooled WMD: 8.54, 95% CI 1.22 to 15.85), multimodal treatment (pooled WMD: 8.21, 95% CI 1.22 to 15.19), mobilisation (pooled WMD: 7.34, 95% CI 0.77 to 13.90) were significantly effective. The SUCRA results showed that manipulation (SUCRA=78.2%) had the highest probability of being the best treatment option (online supplemental figure 6). The quality of this evidence was judged to be moderate to very low (online supplemental table 14).

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Table 1 Comparative effectiveness for neck pain intensity

Multimodal treatment						
−0.03 (−24.65, 24.59)	Active control					
−0.80 (−23.26, 21.66)	−0.77 (−25.04, 23.49)	Manual therapy				
−7.93 (−30.30, 14.44)	−7.90 (−32.08, 16.28)	−7.13 (−29.10, 14.85)	Soft tissue technique			
−28.82 (−55.11, −2.53)	−0.90 (−16.37, 14.57)	−8.03 (−23.63, 7.58)	4.73 (−17.72, 27.19)	Manipulation		
−16.42 (−34.12, 1.29)	−16.39 (−36.33, 3.56)	−15.61 (−32.82, 1.59)	−8.49 (−25.57, 8.60)	−5.82 (−29.71, 18.08)	Mobilisation	
−36.65 (−61.02, −12.28)	**−36.62 (−62.67, −10.57)**	**−35.85 (−59.86, −11.83)**	**−28.72 (−52.65, −4.79)**	16.61 (−6.40, 39.63)	**−20.23 (−39.87, −0.60)**	No treatment

Note: Estimates are presented as column versus row for the network meta-analyses. Effect estimates are presented as (pooled) weighted mean difference (WMD) with 95% confidence intervals. (Pooled) WMD<0 represents the superiority of the column-defining treatment in the network evidence; (pooled) WMD>0 indicates the superiority of row-defining treatments in the network evidence. Statistically significant results are in bold. Details of the certainty of evidence are provided in online supplemental tables 6 and 7).
Certainty of evidence: Moderate Low Very low.

Subgroup analysis

Subgroup analyses stratified by the total scale range of the primary outcome (100-point vs 10-point). For subgroup 1, a total of seven interventions from 14 trials were included, with evidence of global inconsistency (p<0.0001). For subgroup 2, a total of 7 interventions from 54 trials were included, with evidence of global inconsistency (p=0.0433). The result showed that multimodal treatment was observed to have consistent superiority across both subgroups for neck pain intensity reduction. Discrepancies in intervention rankings were observed both between subgroups and relative to the primary analysis, suggesting that differential sensitivity of pain measurement scales may underlie this instability (online supplemental figures 7–10); (online supplemental tables 15–21).

Sensitivity analysis

A sensitivity analysis was conducted on neck pain intensity by excluding RCTs with large differences in baseline scores. A total of 7 interventions from 61 trials were included, with evidence of global inconsistency (p=0.0485). Sensitivity analyses produced results consistent with the primary analysis, showing only slight variations in effect estimates and no changes in the ranking of intervention effectiveness, supporting the robustness of the main findings (online supplemental figures 11 and 12); (online supplemental tables 22–26).

Publication bias

For neck pain intensity, Egger's test indicated no evidence of publication bias, with p=0.2436 in the primary analysis and p=0.8264 in the sensitivity analyses. For subgroup analysis, Egger's test indicated no evidence of publication bias, with p=0.9471 for subgroup 1 and p=0.5147 for subgroup 2. In contrast, significant publication bias was observed for neck disability, cervical range of flexion and extension, with p=0.0400, p=0.0071 and p=0.0048, respectively (online supplemental figures 13–19).

DISCUSSION

In this systematic review, we compared the effectiveness of musculoskeletal manipulations, including 7 interventions, from 101 RCTs. Our findings indicate that multimodal treatment was the most effective for improving neck pain intensity and neck disability. Furthermore, mobilisation was potentially the most effective for improving flexion and extension.

Based on the results of both pairwise meta-analysis and NMA, different types of musculoskeletal manipulations, used either individually or in combination, effectively reduced the intensity of neck pain. The NMA results showed that multimodal treatment, active control, manual therapy and soft tissue techniques all achieved clinically meaningful improvements, with their respective WMDs exceeding the established threshold of 20 points. These interventions demonstrated comparable effect sizes, suggesting that each modality may offer similar

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therapeutic benefits. Notably, multimodal treatment had the best clinical effect (SUCRA=100%), which was consistent with clinical observations and previous study findings.[16 37 105] Multimodal treatment is likely to achieve superior effects through simultaneous targeting of muscular hypertonicity and neural sensitisation, whereas single techniques address isolated components.[106–108]

Regarding neck disability, although the pairwise meta-analysis showed that only manual therapy demonstrated a statistically significant effect among the five musculoskeletal manipulations, its effect did not reach the threshold for clinical significance. In contrast, the NMA demonstrated that all interventions were effective, with multimodal treatment and mobilisation achieving clinically meaningful improvements (WMDs>15 points). The highest SUCRA ranking supports multimodal treatment (SUCRA=96.3%) as the most effective intervention. Musculoskeletal manipulations may improve neck disability by restoring joint kinematics, while active exercise can promote motor relearning through cortical engagement.[109–112] Their combination may lead to sustained functional gains via neuromuscular adaptation. Although the presence of publication bias for neck disability may have led to an overestimation of effect sizes, multimodal treatment, as the most effective intervention for both neck pain and disability, should still be recommended in clinical practice.

Based on the results of pairwise meta-analysis, manipulation, mobilisation and soft tissue techniques were effective in improving cervical range of motion (both flexion and extension). However, the NMA identified manipulation, multimodal treatment and soft tissue techniques as effective, with manipulation being the most effective (SUCRA=81.3% for flexion and SUCRA=78.2% for extension). Although none of the interventions achieved the threshold for clinical significance (10 degree), this finding suggests that directly targeting the cervical vertebral body may enhance cervical mobility.[113 114] This may be achieved by restoring arthrokinematics through mechanotransductive effects on joint capsules and periarticular tissues. Based on the funnel plot, our NMA findings may be influenced by publication bias. Future studies should be included to obtain more robust and conclusive results.

The quality of evidence from our results was moderate to very low because of the unsatisfactory methodological quality of the included studies. Specifically, 80 RCTs (79.2%) were judged to have some concerns or a high risk of bias and 80 RCTs (79.2%) had a sample size of fewer than 100 participants. This finding indicated that the effect estimates changed by including further RCTs. Regarding the risk of bias assessment, 45 RCTs (44.5%) did not report information on blinding of outcome assessors, and 52 RCTs (51.4%) did not provide protocol registration details. As a result, more than 50% of the included RCTs were rated as having some concerns or high risk of bias. Therefore, future RCTs should improve the implementation and reporting of outcome assessor blinding and provide prior protocols. Based on the reported information, musculoskeletal manipulations are associated with minor and transient adverse events, such as tiredness, muscular tension and aggravated pain.[115] However, only five included RCTs reported adverse events[116]; thus, clinicians should carefully consider the effectiveness, certainty of evidence and safety of interventions during clinical decision-making.

Some strengths of this review included a comprehensive systematic review methodology that adhered to the Cochrane Handbook and PRISMA-NMA reporting guidelines. Different musculoskeletal manipulations were compared, providing vital information for physicians. This study had some limitations. First, the inability to conduct a subgroup analysis based on neck pain timelines (eg, acute and chronic) due to a lack of relevant information represents a critical gap, as differential treatment responses may exist across these populations. Second, only a few studies assessed long-term persistent effects through follow-up; consequently, our findings primarily reflect short-term outcomes, and the comparative sustainability of interventions remains unknown. Third, to avoid a sparse network, we did not separate the no-treatment and waiting list from sham and/or placebo interventions because only a few studies reported a no-treatment or waiting list approach.

CONCLUSIONS

Our findings indicate that multimodal treatment, the soft tissue technique, manipulation, manual therapy and mobilisation effectively alleviate the intensity of neck pain and neck disability, with multimodal treatment being the optimal choice. This could assist in clinical decision-making. However, owing to the limited quality of the RCTs included, clinicians should carefully consider the effectiveness of these approaches and the certainty of evidence.

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